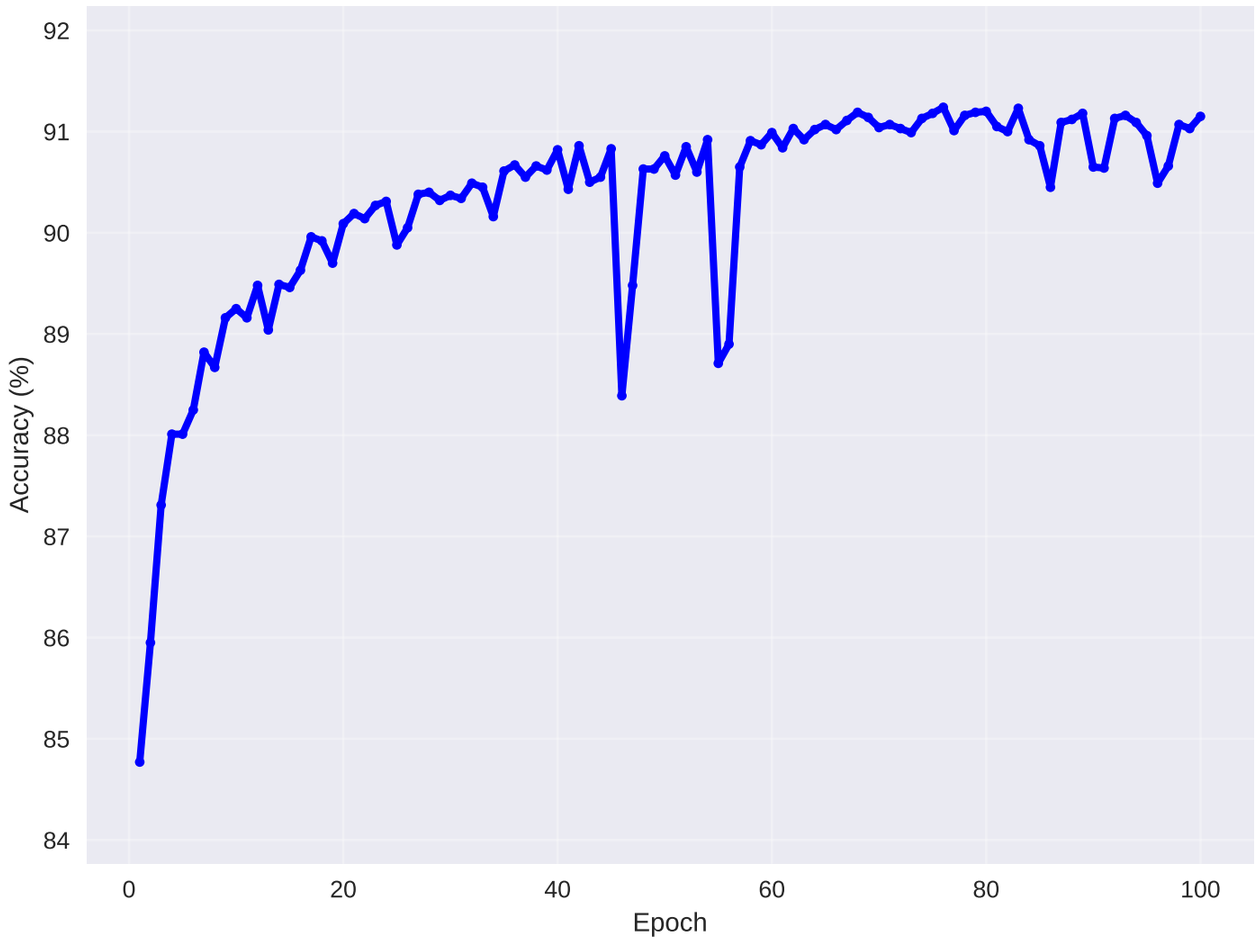
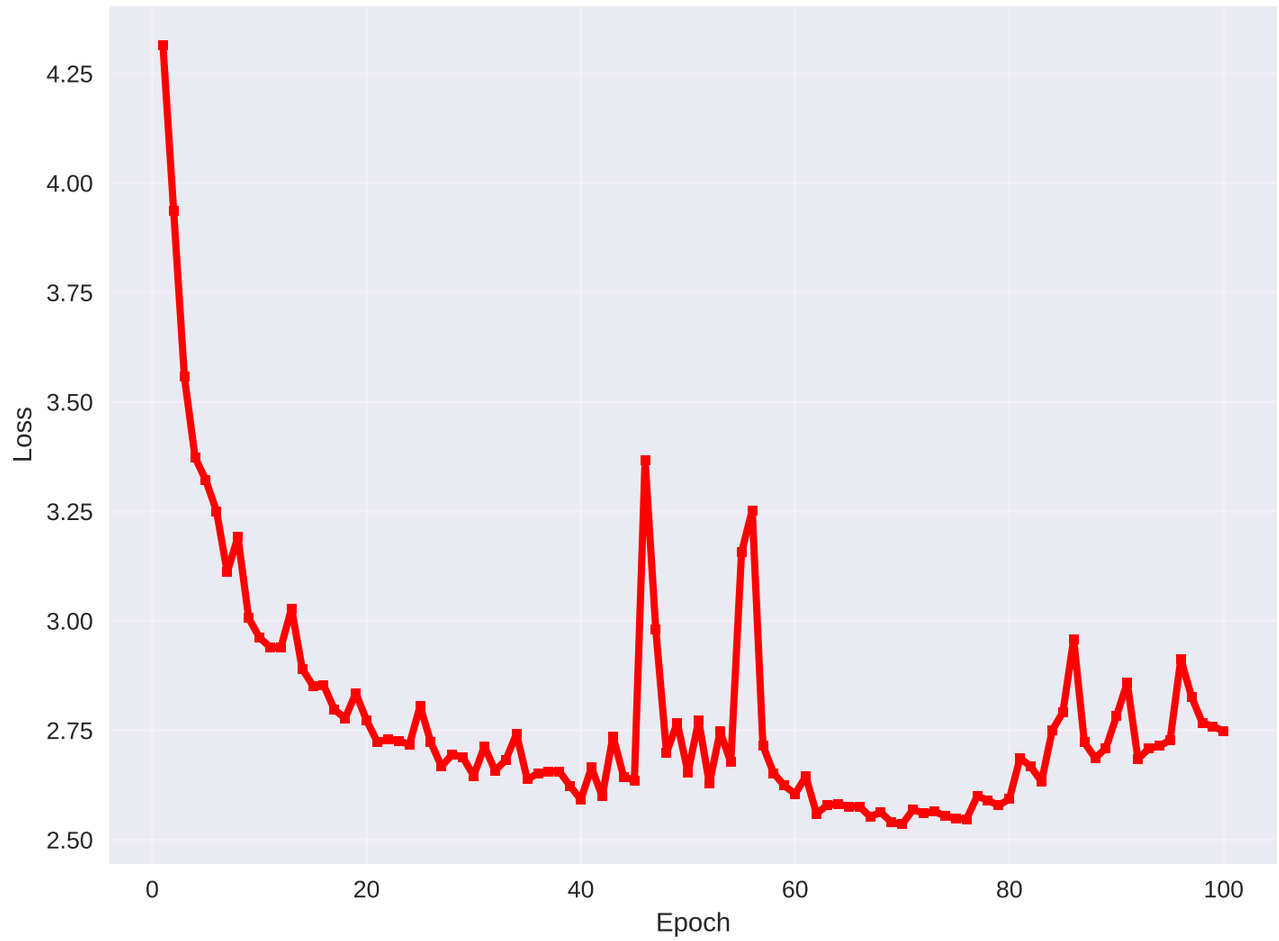
Federated Learning - Main Metrics Overview

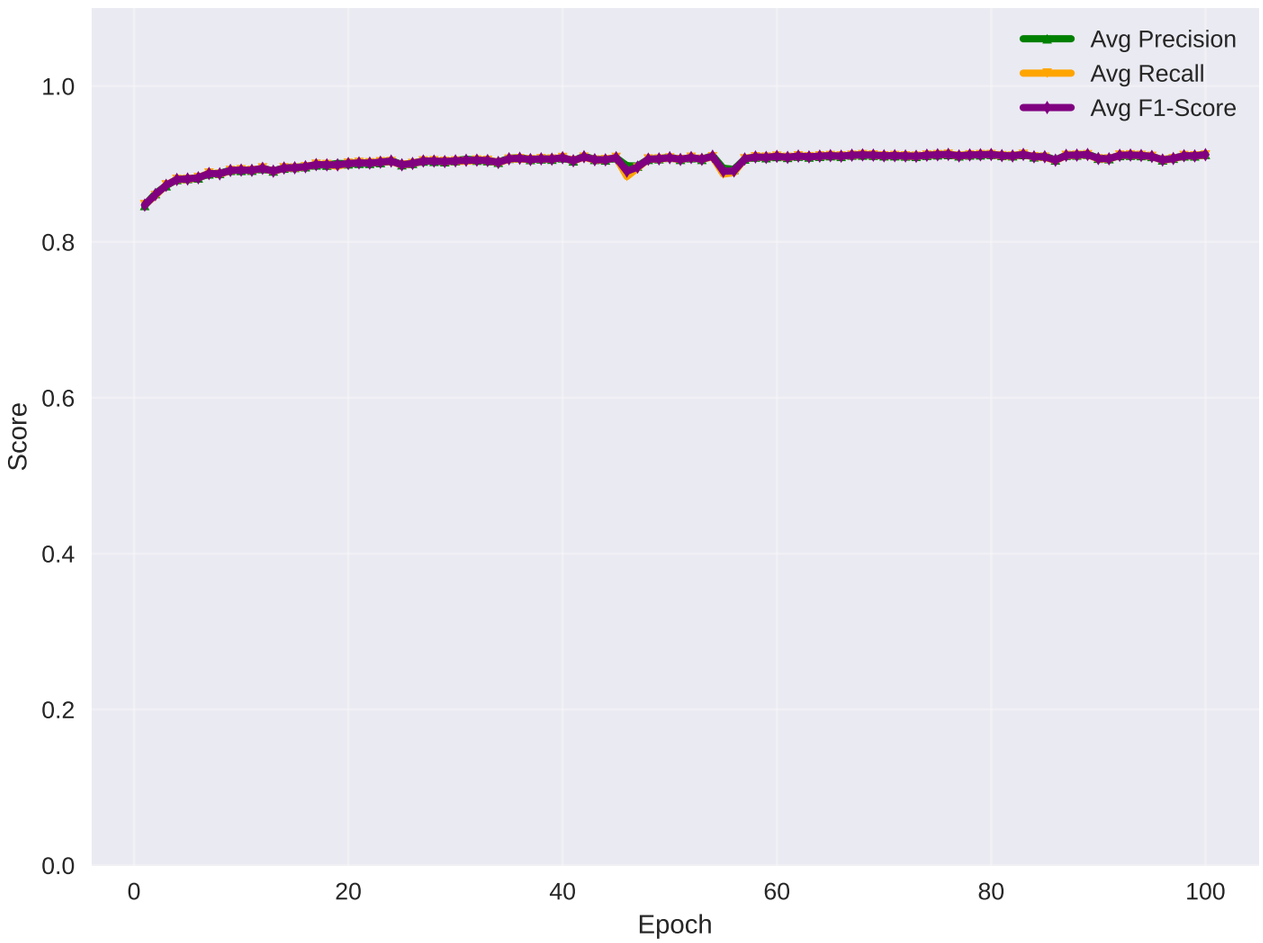
 Global Model Accuracy Evolution



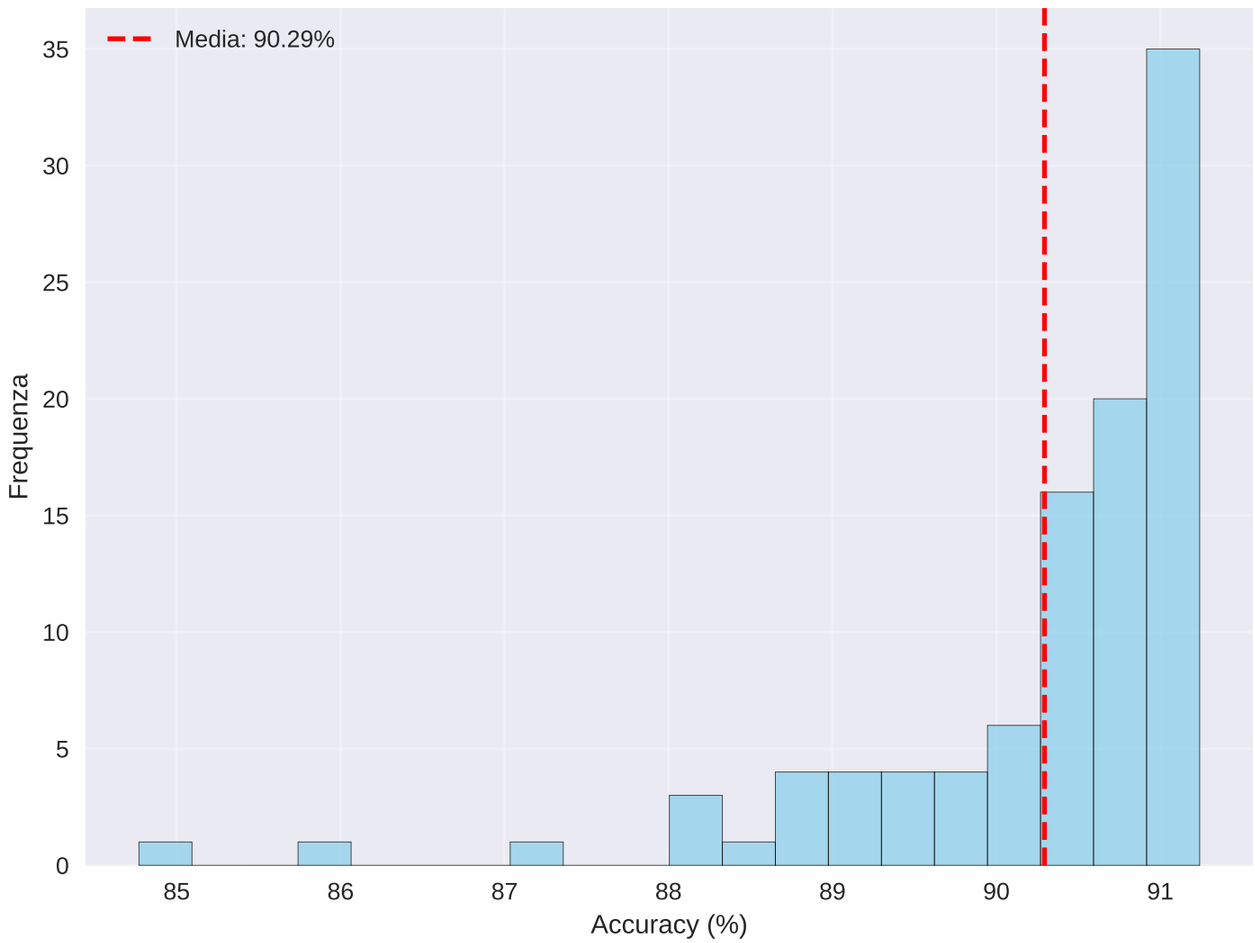
 Global Model Loss Evolution



 Average Precision, Recall & F1-Score

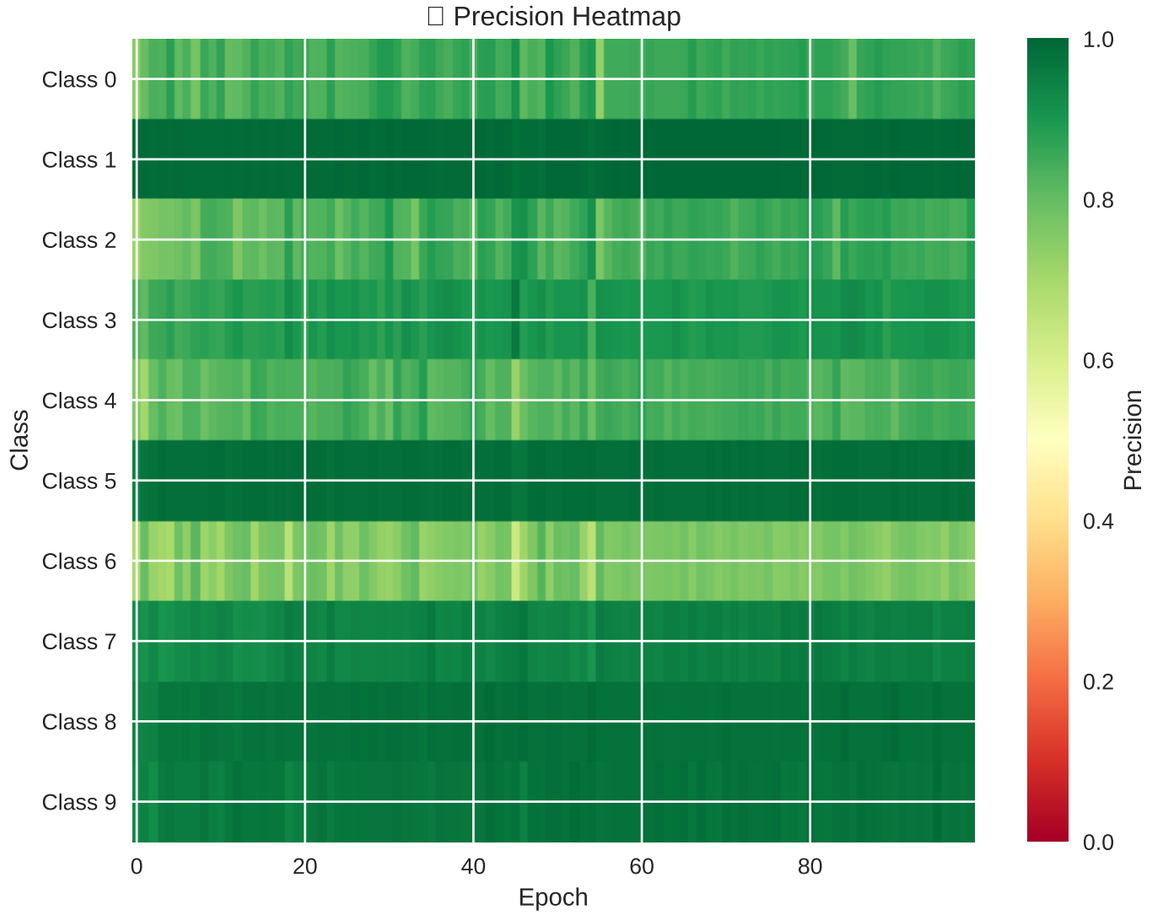


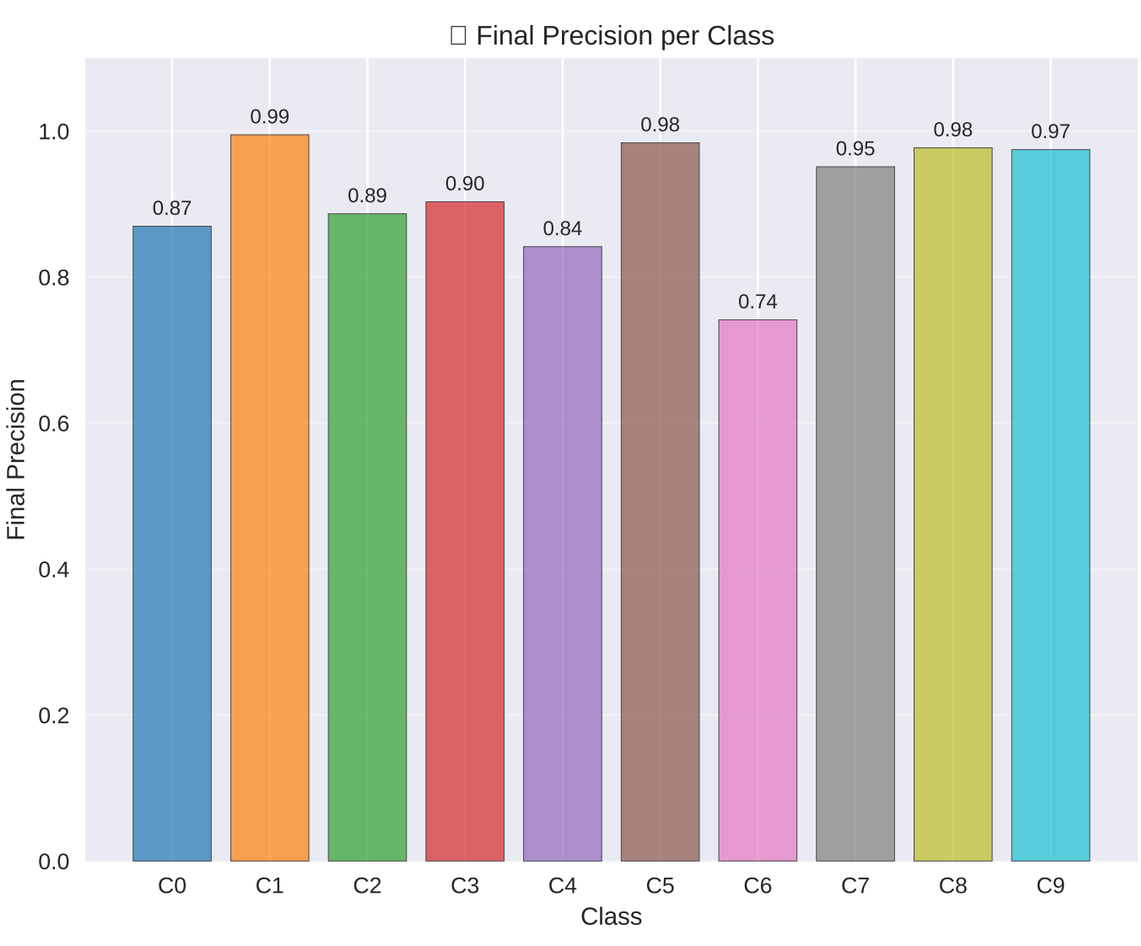
 Distribuzione Accuracy

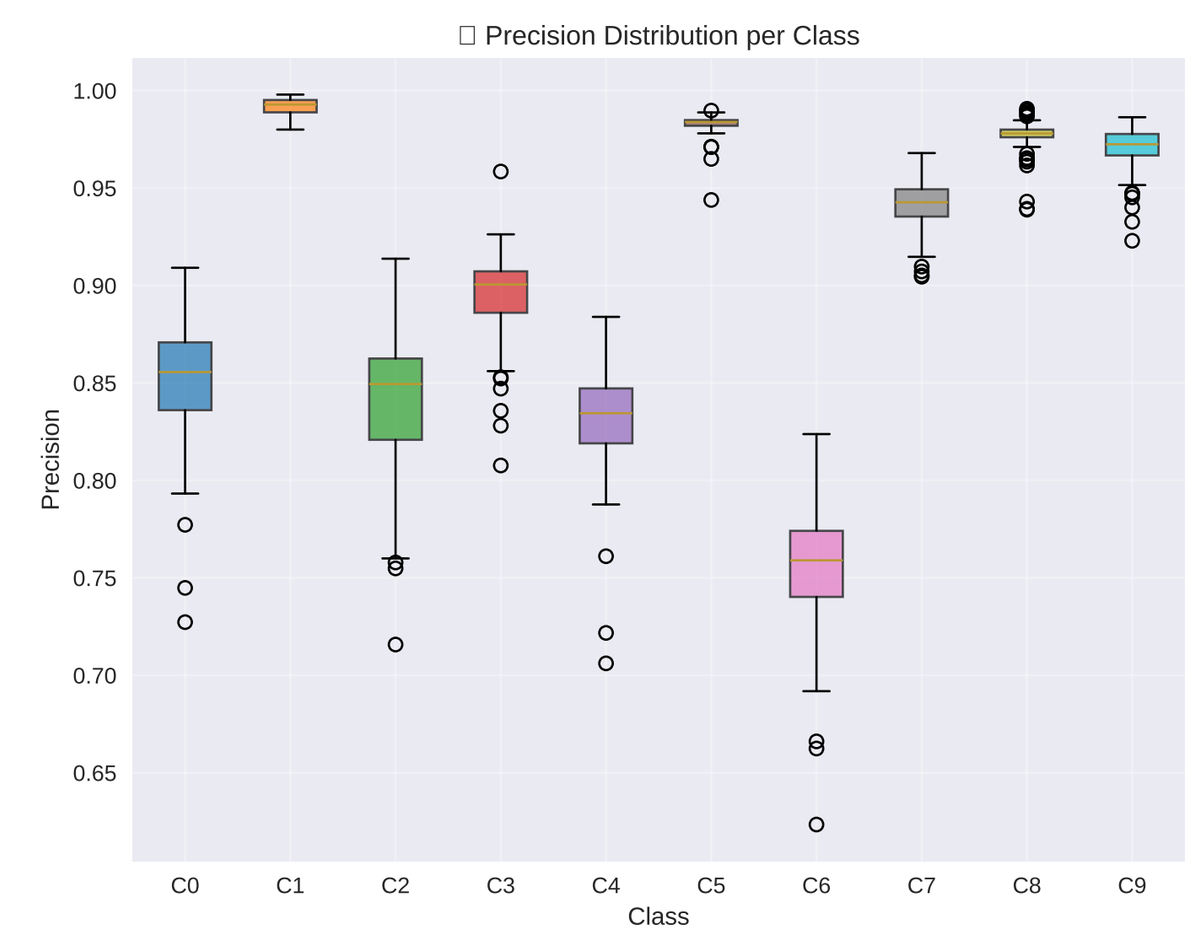


Per-Class Precision Analysis



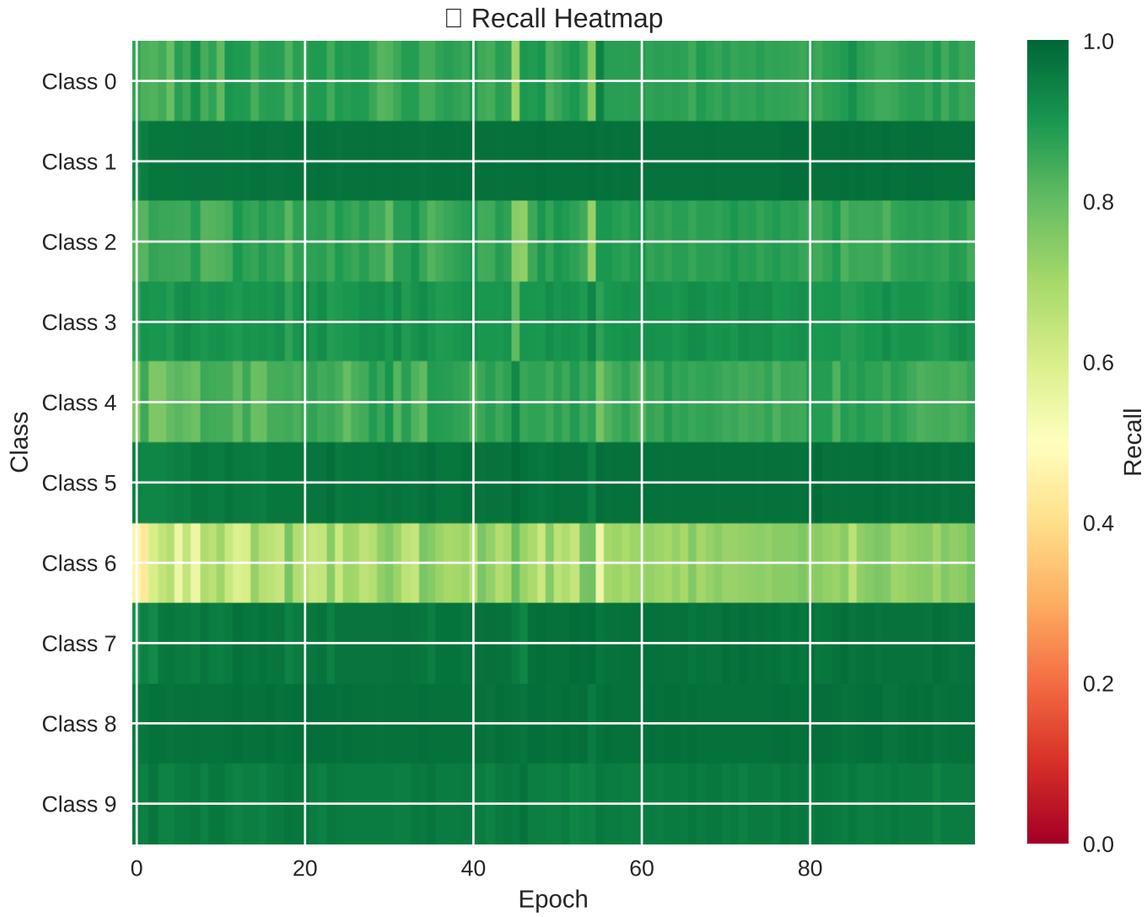


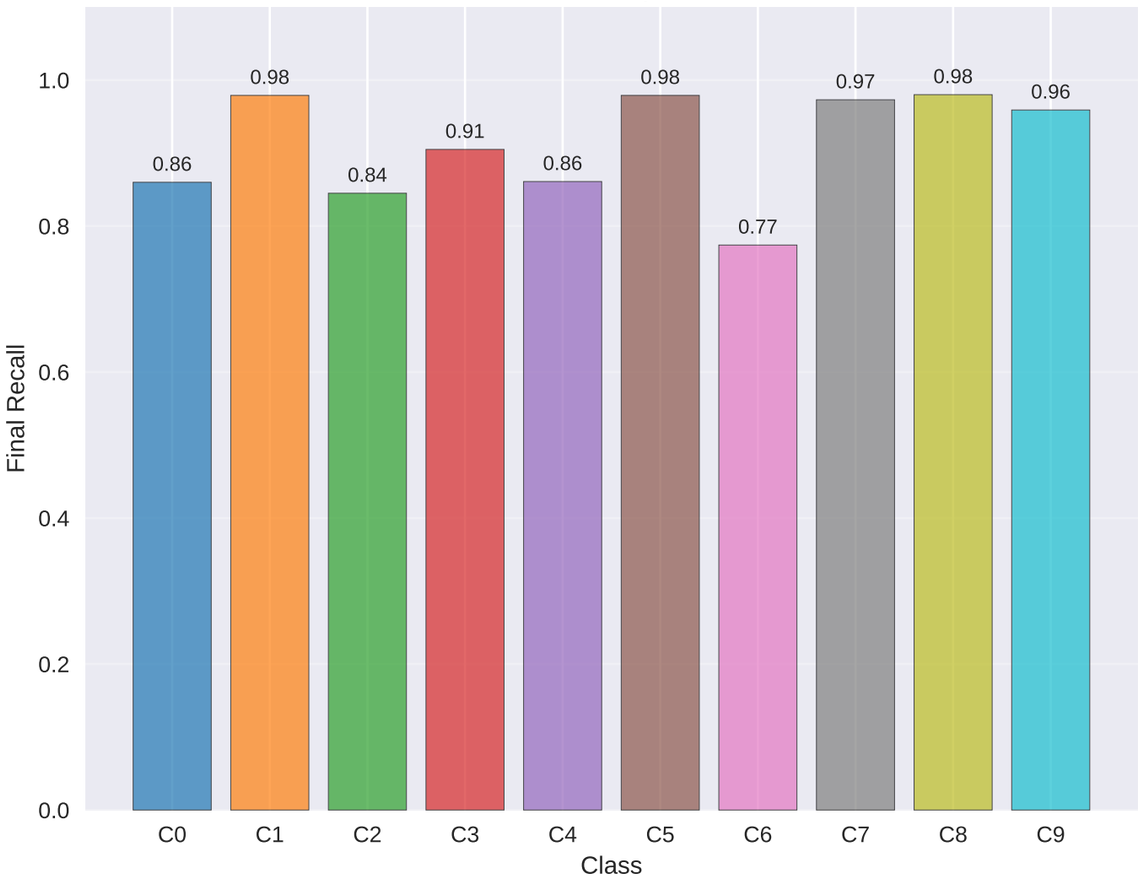


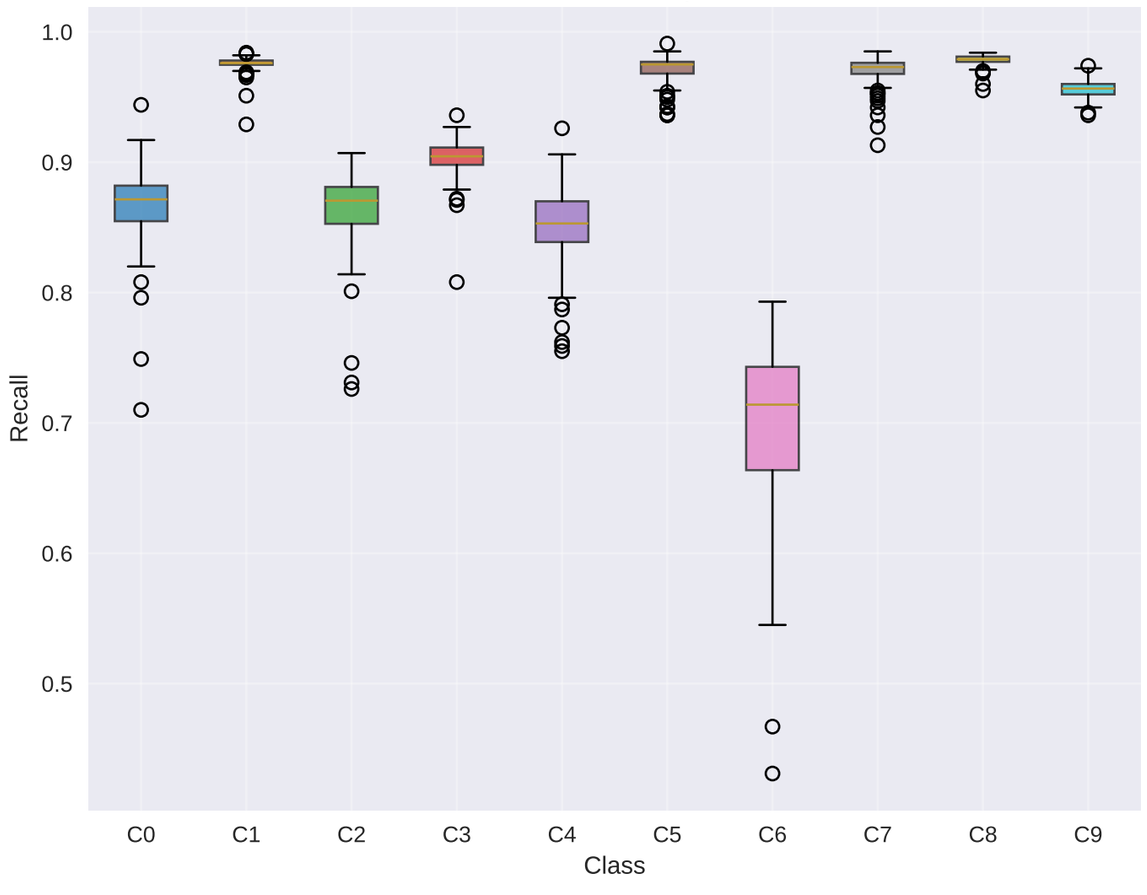


Per-Class Recall Analysis

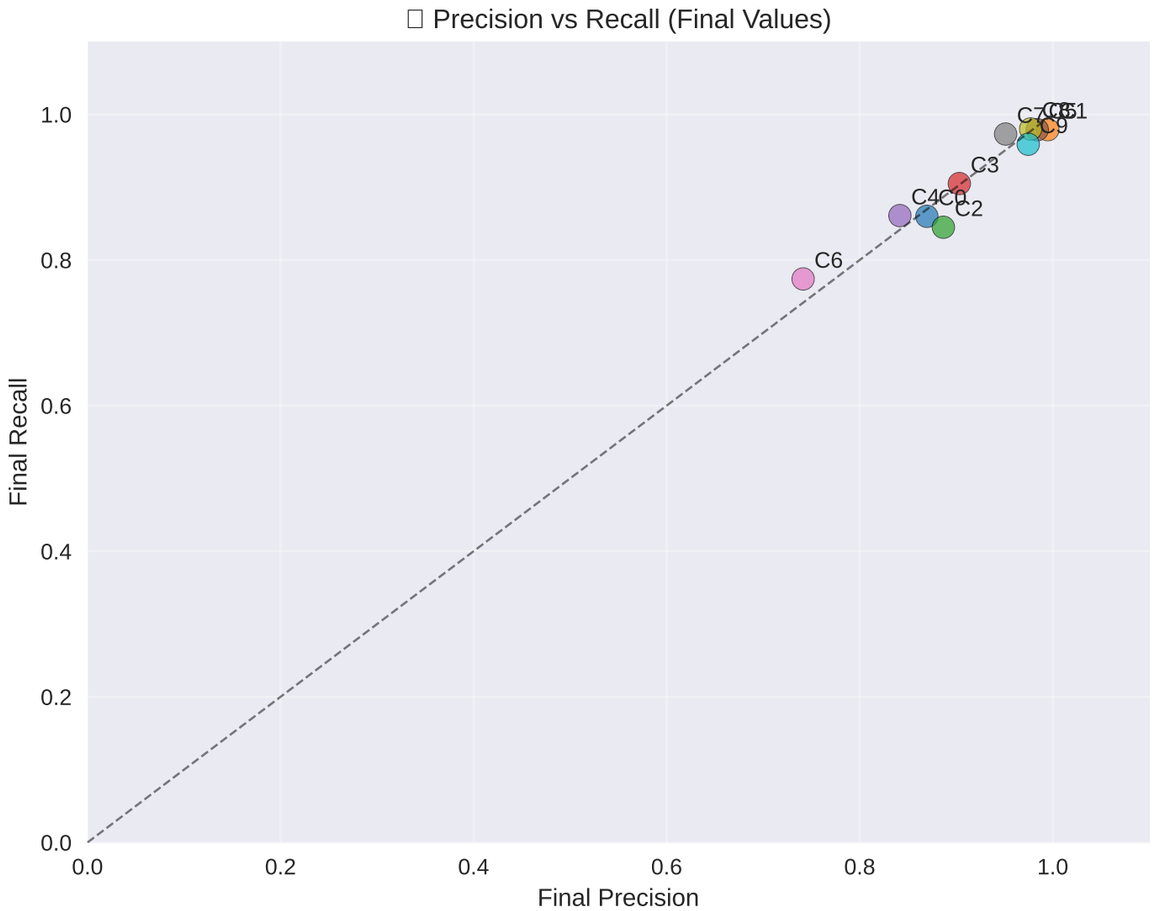




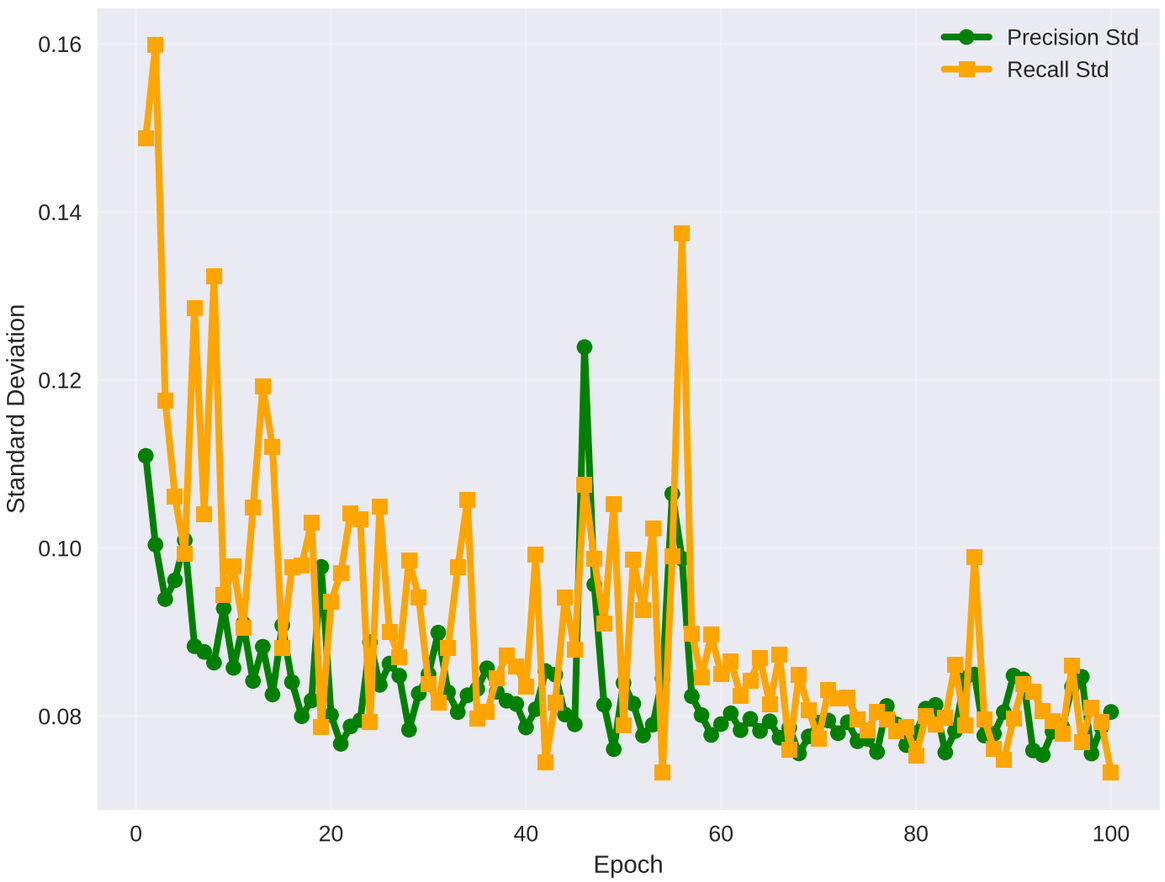


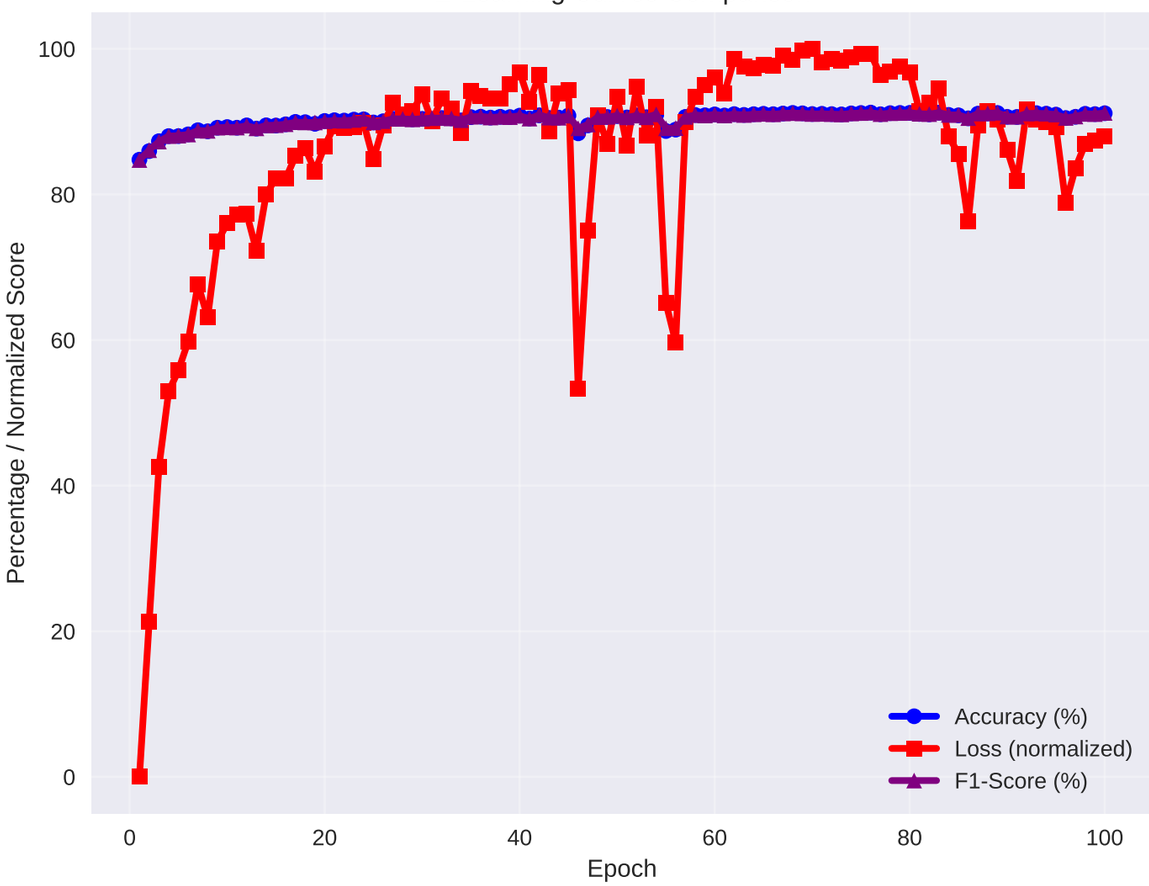


Comparative Analysis









Performance Summary Report

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FEDERATED LEARNING - PERFORMANCE SUMMARY
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FILE ANALIZZATO: 3000_results.csv
DATA ANALISI: 2025-08-29 22:40:51
EPOCHE TOTALI: 100

METRICHE FINALI:
• Accuracy: 91.15%
• Loss: 2.7485
• Average Precision: 0.912
• Average Recall: 0.911
• Average F1-Score: 0.912

EVOLUZIONE TRAINING:
• Miglioramento Accuracy: +6.38%
• Riduzione Loss: +1.5665
• Stabilità Accuracy (CV): 0.0123

PERFORMANCE PER CLASSE:
• Migliore Precision: Classe 1 (0.995)
• Peggiore Precision: Classe 6 (0.741)
• Migliore Recall: Classe 8 (0.980)
• Peggiore Recall: Classe 6 (0.774)

ANALISI DETTAGLIATA PER CLASSE:
• Classe 0: P=0.870, R=0.860, F1=0.865
• Classe 1: P=0.995, R=0.979, F1=0.987
• Classe 2: P=0.887, R=0.845, F1=0.865
• Classe 3: P=0.903, R=0.905, F1=0.904
• Classe 4: P=0.842, R=0.861, F1=0.851
• Classe 5: P=0.984, R=0.979, F1=0.981
• Classe 6: P=0.741, R=0.774, F1=0.757
• Classe 7: P=0.951, R=0.973, F1=0.962
• Classe 8: P=0.977, R=0.980, F1=0.979
• Classe 9: P=0.975, R=0.959, F1=0.967

VALUTAZIONE QUALITATIVA:
• Performance Globale: BUONA
• Stabilità Training: STABILE
• Balance Classes: SBILANCIATO

RACCOMANDAZIONI:
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